

- 3 Female *Anopheles* mosquitoes are the vectors that transmit malaria. Use of insecticides to control the transmission of malaria has resulted in many *Anopheles* mosquitoes becoming insecticide-resistant.

The fungus *Metarhizium pingshaense* infects and kills mosquitoes. Some biologists investigated how this fungus could be used to prevent the transmission of malaria by *Anopheles* mosquitoes. The biologists also genetically modified *M. pingshaense* to produce a toxin that is usually made by spiders.

- (a) The biologists carried out a laboratory procedure to investigate the effect of *M. pingshaense* on the reproduction of *Anopheles coluzzii* mosquitoes. The percentage of female *A. coluzzii* mosquitoes that produced eggs was determined for:

- a control group of uninfected *A. coluzzii* mosquitoes
- *A. coluzzii* mosquitoes infected with *M. pingshaense* that had **not** been genetically modified (non-GM)
- *A. coluzzii* mosquitoes infected with *M. pingshaense* that had been genetically modified (GM).

Figure 3.1 shows the results of the investigation. Error bars show 95% confidence intervals (95% CI).

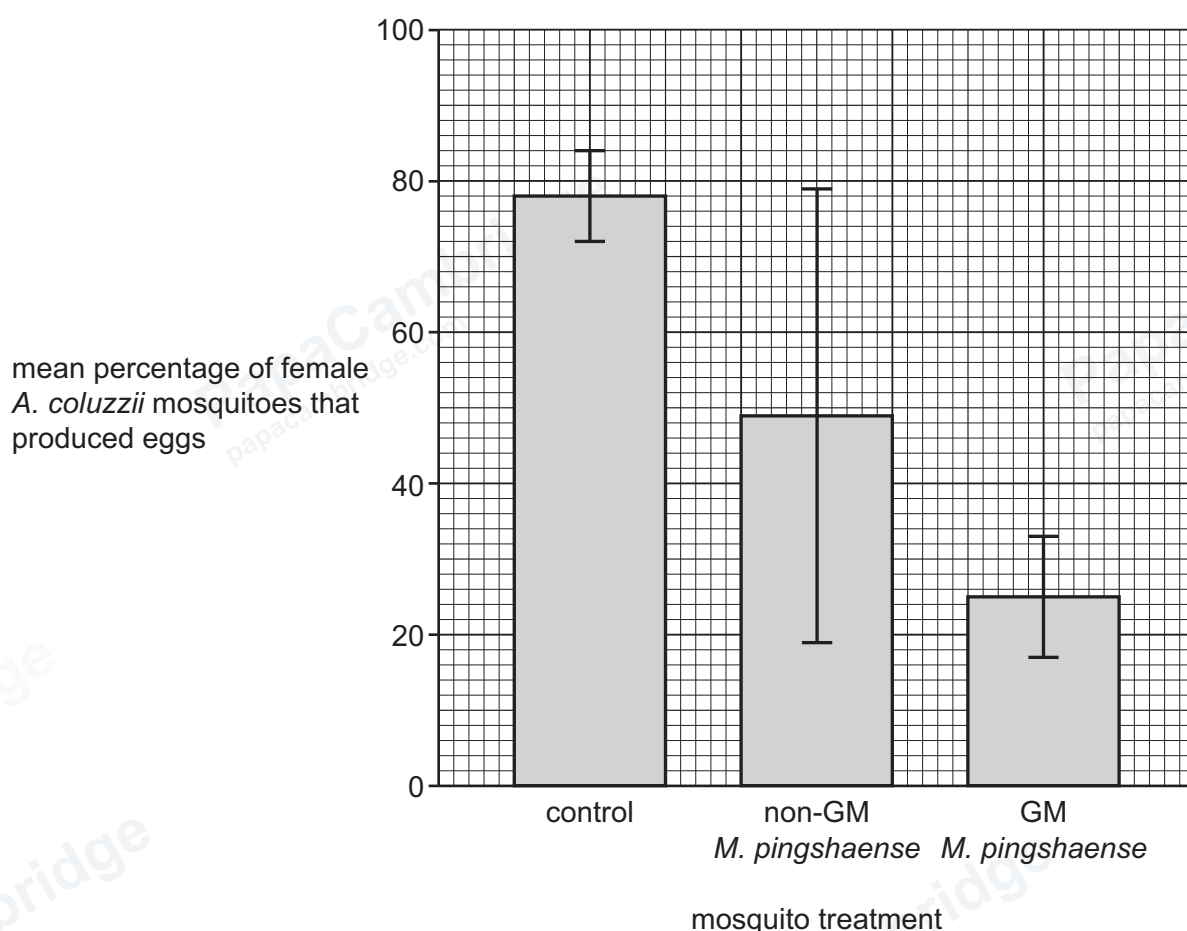


Figure 3.1



Use Figure 3.1 to outline what can be concluded from the data.

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(b) Malaria is a common infectious disease in Burkina Faso, a country in west Africa.

The biologists investigated the effect of GM *M. pingshaense* on the reproduction of *A. coluzzii* mosquitoes in a natural environment in Burkina Faso.

The biologists set up two 9.55 m × 9.10 m experimental areas in Burkina Faso in 2017. With the help of people from Burkina Faso, the biologists:

- built a traditional hut inside each experimental area
- planted fruit trees as a source of sugar for the mosquitoes to feed on
- made small ponds as a source of water for the reproduction of mosquitoes.

Figure 3.2 shows some people building a traditional hut in Burkina Faso.



Figure 3.2

This procedure was used in one of the experimental areas.

- A piece of black cotton fabric was placed over one wall of the hut.
- GM *M. pingshaense* was added to the piece of black cotton fabric. The *A. coluzzii* mosquitoes were attracted to the black cotton fabric.
- 1000 male *A. coluzzii* mosquitoes and 500 female *A. coluzzii* mosquitoes were released into the hut.
- Some livestock were placed inside the hut every evening. Female *A. coluzzii* mosquitoes fed on blood from the livestock. This did not harm the livestock.
- The number of *A. coluzzii* mosquitoes was determined every day for 45 days. Two generations of *A. coluzzii* mosquitoes were produced in the 45 days.

The procedure was also carried out in the other experimental area, but *M. pingshaense* was **not** added to the piece of black cotton fabric.

The investigation was carried out in the summer, when rainfall is high in Burkina Faso (rainy season).

Three repeats of the whole investigation were carried out, each lasting for 45 days. The amount of rainfall decreased during repeat 2 and repeat 3.

- (i) Suggest a method the biologists could use to prevent *A. coluzzii* mosquitoes escaping from the experimental areas.

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- (ii) The scientists used data from the investigation to calculate the total number of *A. coluzzii* mosquitoes in the second generation.

Table 3.1 shows the results of the investigation.

Table 3.1

repeat	total number of <i>A. coluzzii</i> mosquitoes in the second generation	
	experimental area with GM <i>M. pingshaense</i>	experimental area with no <i>M. pingshaense</i>
1	13	1396
2	61	731
3	12	429

A student used the chi-squared test (χ^2) to analyse the data from the investigation.

The student stated the null hypothesis as:

there is no difference between the number of *A. coluzzii* mosquitoes in the second generation from the experimental area with GM *M. pingshaense* and from the experimental area with no *M. pingshaense*.

The equation for the calculation of χ^2 is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

key

O = observed result

E = expected result

Σ = sum of

Complete Table 3.2 to calculate the value of χ^2 for the data from the investigation.

Give the value of χ^2 to **three** significant figures.

Table 3.2

repeat	O	E	$O - E$	$(O - E)^2$	$\frac{(O - E)^2}{E}$
1	13	1396			
2	61	731			
3	12	429			
					$\chi^2 = \dots\dots\dots$

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- (iii) The student compared the calculated value of χ^2 to the critical values shown in Table 3.3.

Table 3.3

degrees of freedom	probability level (p)			
	0.50	0.10	0.05	0.01
1	0.455	2.706	3.841	6.635
2	1.386	4.605	5.991	9.210
3	2.366	6.251	7.815	11.345
4	3.357	7.779	9.488	13.277

The student rejected the null hypothesis.

Use Table 3.3 to identify the critical value for this investigation **and** explain whether the student was correct to reject the null hypothesis.

critical value

explanation

[1]

(c) The student concluded that people in Burkina Faso could use pieces of black cotton fabric with GM *M. pingshaense* inside their houses to reduce the transmission of malaria.

Explain how the data in Table 3.1 and the calculated value of χ^2 support this conclusion.

Suggest what further information would be needed before this method could be used to reduce the transmission of malaria in Burkina Faso.

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[Total: 11]